

24Bit, 96KHz Sampling Enhanced Multi-Bit Delta-Sigma Audio DAC

Description

This high performance audio CODEC core is designed targeting at high-end portable audio devices. The audio DAC utilize Go2Silicon's enhanced multi-level, delta-sigma architecture that employs multi-order noise shaping and amplitude quantization to achieve excellent SNR performance and a high tolerance to clock jitter. This DAC accepts industry-standard audio data formats with 16- to 24-bit audio data. Sampling rates from 8K up to 96K are supported. The design is in TSMC 0.13um process can also be ported to a different foundry.

Applications

Portable MP3 Players and Recorders

PDA's and Handheld Computers

3G cell phones

LCD TV HDTV DVD

Features

- Stereo 24-bit multi-bit sigma delta DAC are used with over-sampling digital interpolation filters.
- DSP filter algorithm can support two channels DAC timing sharing logic architecture for less gate count
- DAC digital filter can apply digital de-emphasis under register control
- DAC Mute Control & Click and pop suppression
- TSMC 0.13um process
- Scan insertion ready for manufacturing test
- Supports 8K, 8.108k, 11.025K, 12K, 16K, 22.05K, 24K, 32K, 44.1K, 48K, 88.1K and 96K sampling
- Audio performance
 - DAC SNR 97 dB (A weighted)

